

Calculus I

Type 1: Quotient of quadratics.

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Example (Limit with Factoring)

Find $\lim_{x \rightarrow 3} \frac{x^3 - 3x^2 + x - 3}{x^2 - 7x + 12}$

Plug in 3: $\frac{(3)^3 - 3(3)^2 + (3) - 3}{(3)^2 - 7(3) + 12} = -$

Zero over zero is undefined, so we can't use direct substitution.

Factor: $\lim_{x \rightarrow 3} \frac{x^3 - 3x^2 + x - 3}{x^2 - 7x + 12} = \lim_{x \rightarrow 3} \frac{\quad}{\quad}$

$$= \lim_{x \rightarrow 3} \frac{x^2 + 1}{x - 4}$$

Plug in 3: $\lim_{x \rightarrow 3} \frac{x^3 - 3x^2 + x - 3}{x^2 - 7x + 12} = \frac{(3)^2 + 1}{(3) - 4}$

$$= \frac{10}{-1}$$

$$= -10.$$